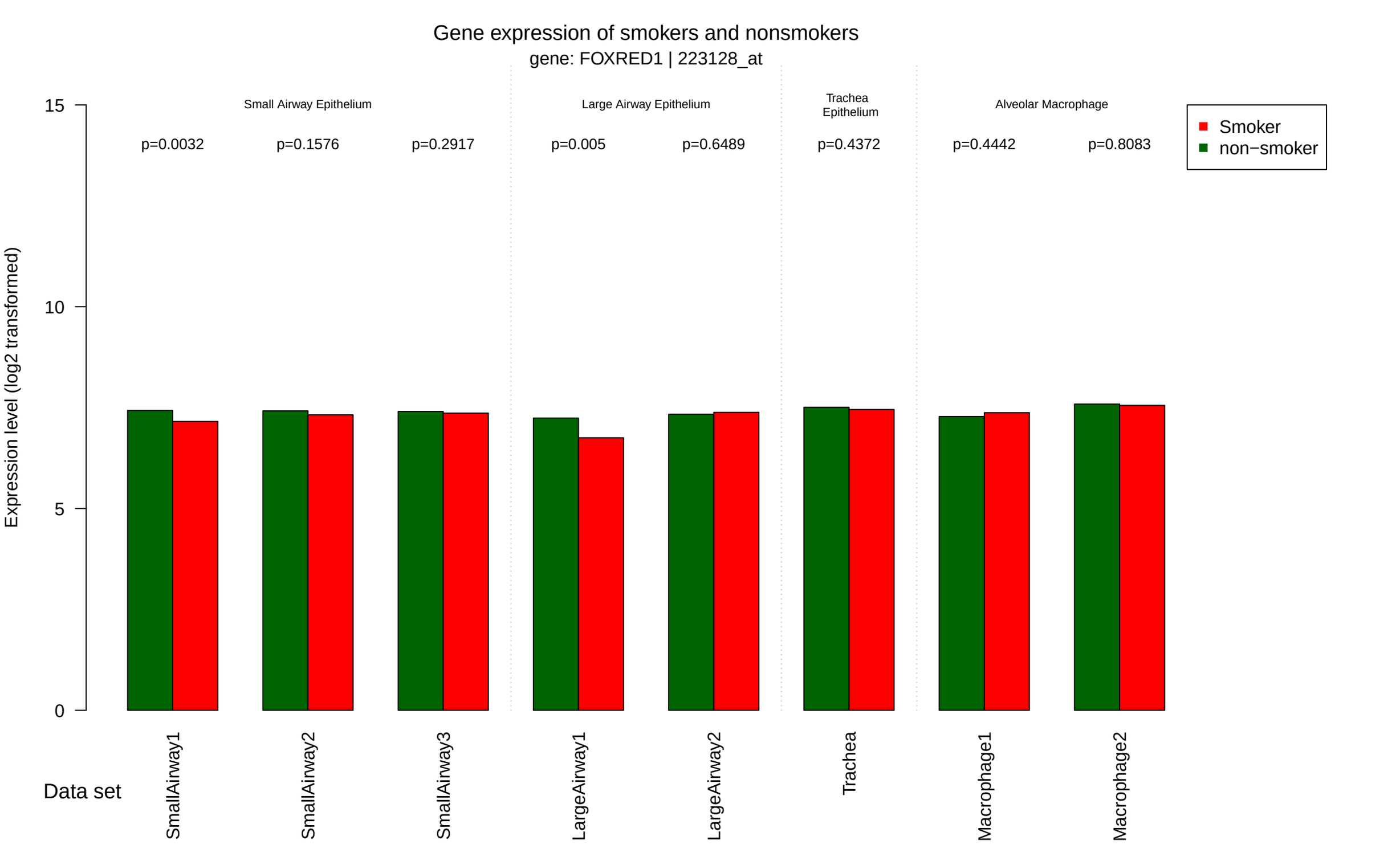
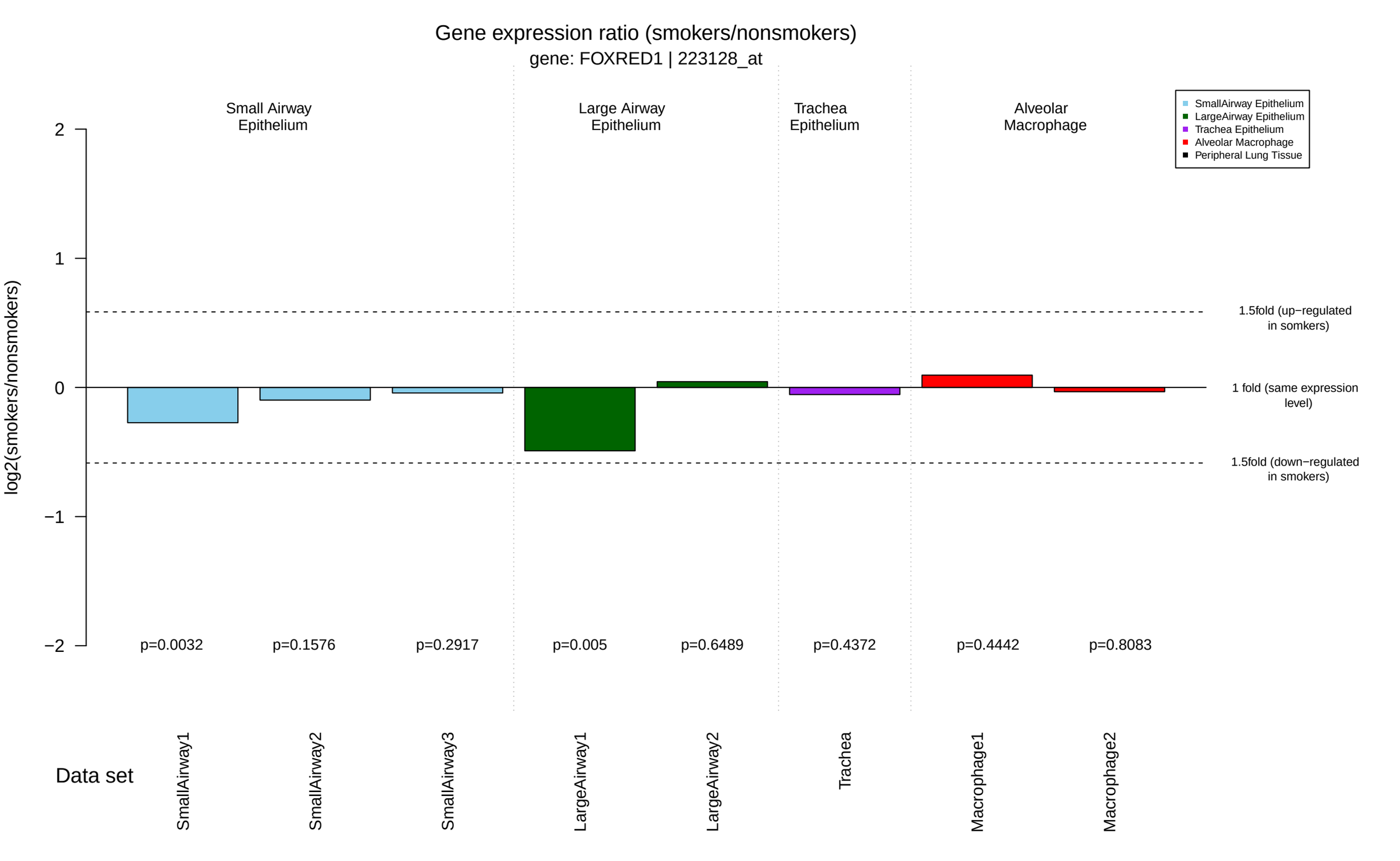
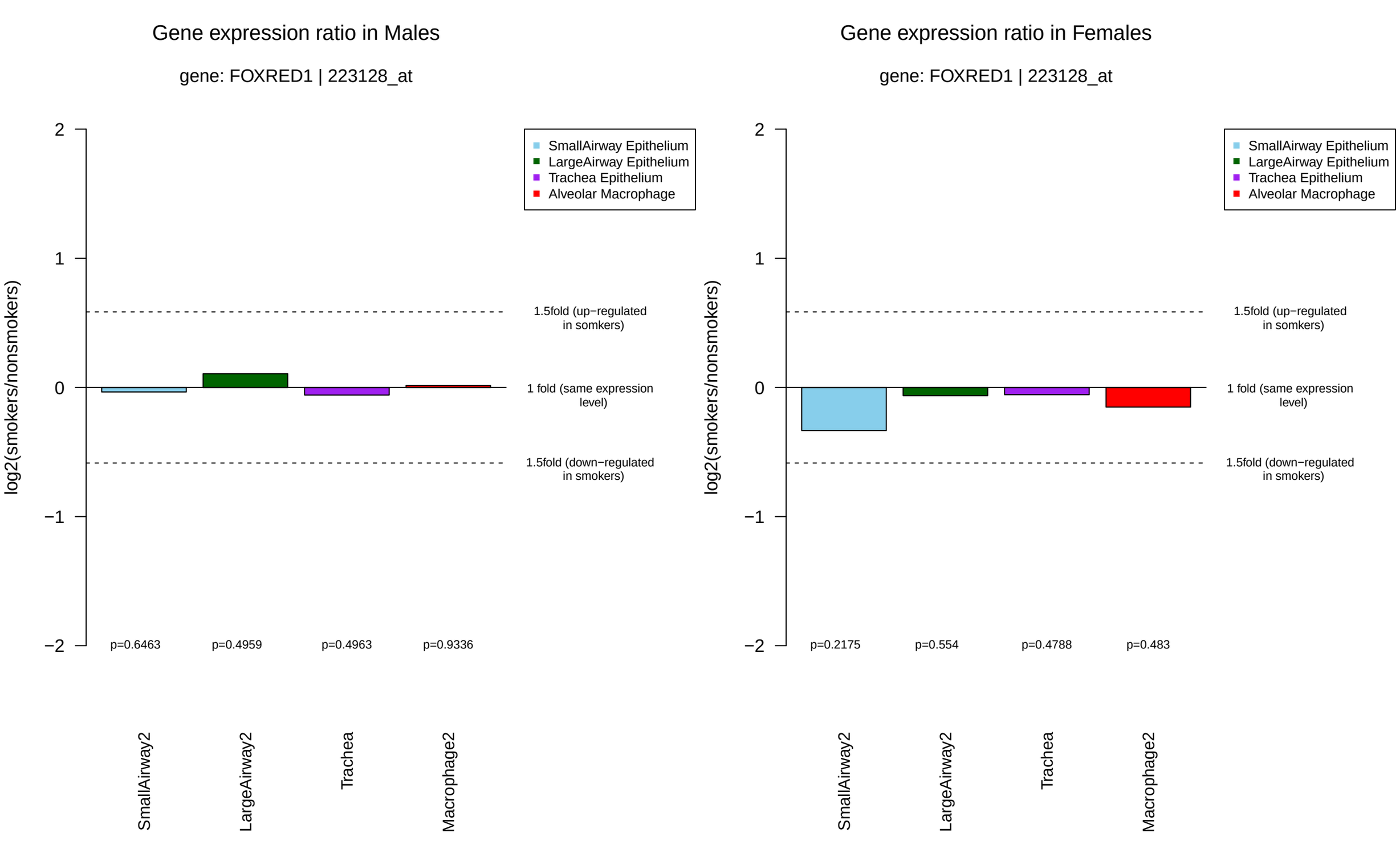
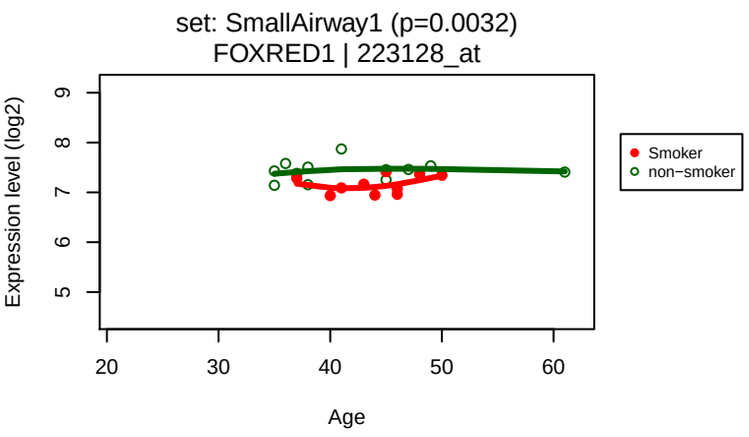
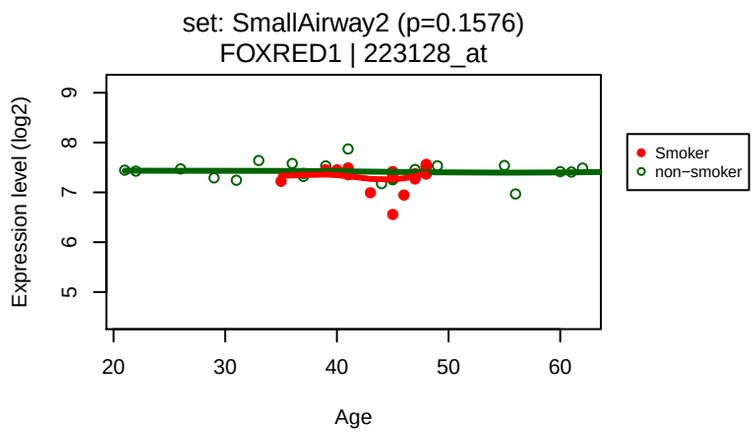
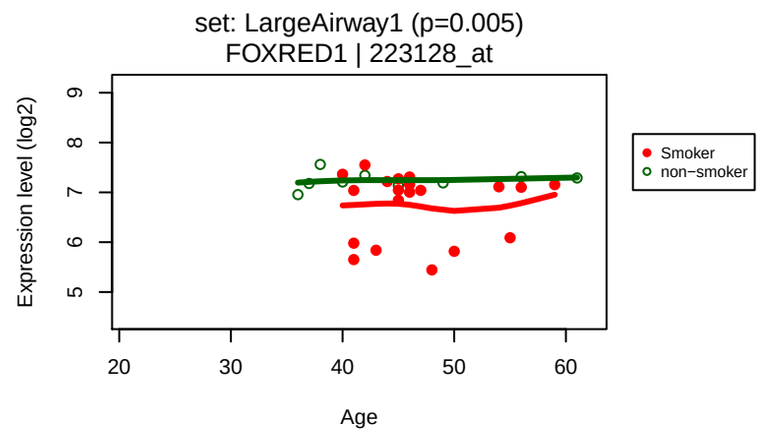

Gene expression of smokers and nonsmokers
gene: FOXRED1 | 223128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0032
p=0.1576
p=0.2917
p=0.005
p=0.6489
p=0.4372
p=0.4442
p=0.8083
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

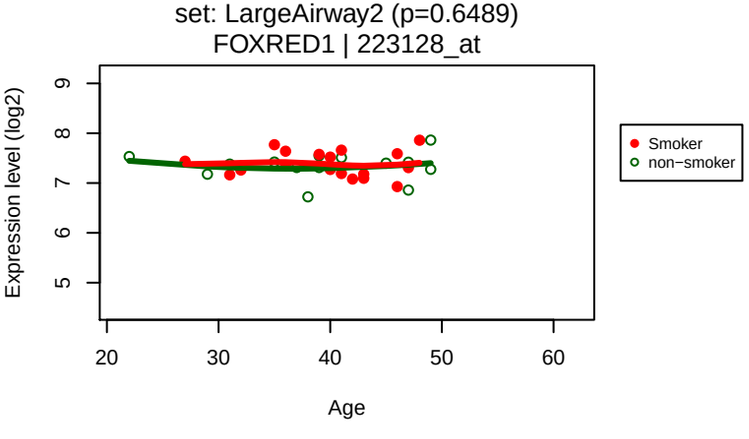

Gene expression ratio (smokers/nonsmokers)
gene: FOXRED1 | 223128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0032
p=0.1576
p=0.2917
p=0.005
p=0.6489
p=0.4372
p=0.4442
p=0.8083
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

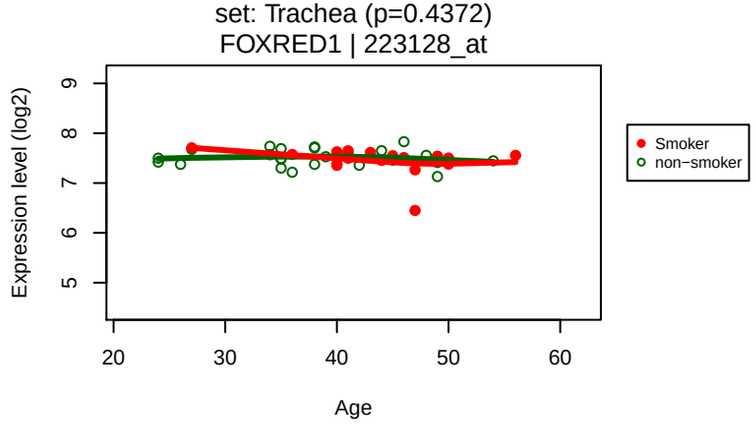

Gene expression ratio in Males
gene: FOXRED1 | 223128_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6463
p=0.4959
p=0.4963
p=0.9336
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FOXRED1 | 223128_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2175
p=0.554
p=0.4788
p=0.483
SmallAirway2
LargeAirway2
Trachea
Macrophage2

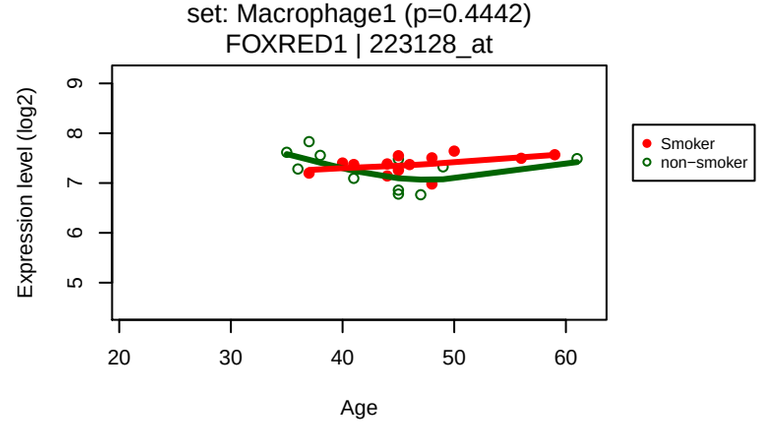

set: SmallAirway1 (p=0.0032)
FOXRED1 | 223128_at
Expression level (log2)
Age
Smoker
non−smoker

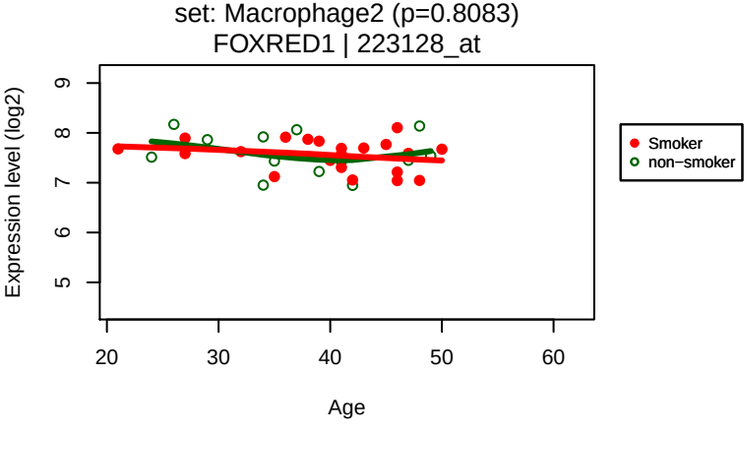

set: SmallAirway2 (p=0.1576)
FOXRED1 | 223128_at
Expression level (log2)
Age
Smoker
non−smoker

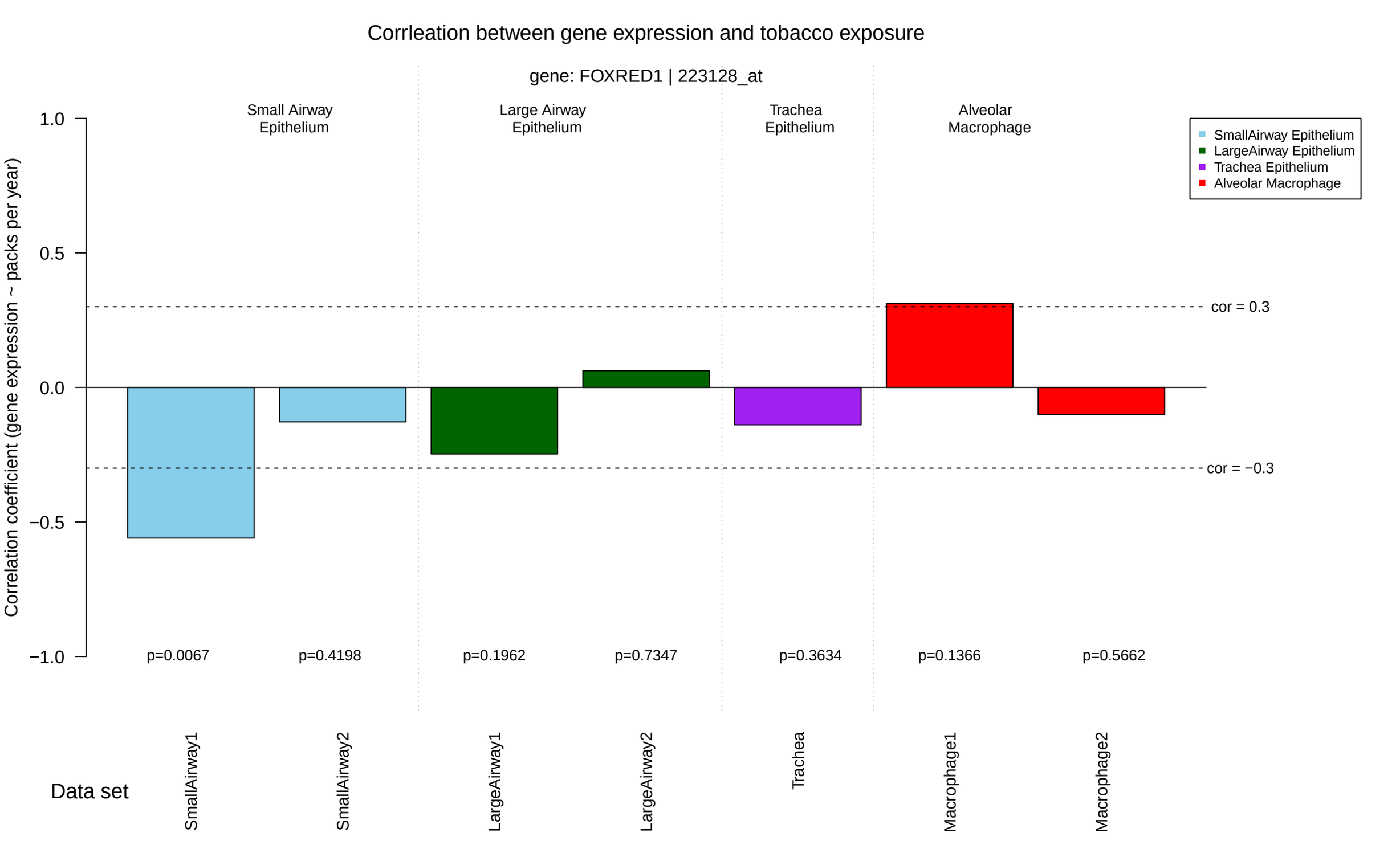

set: LargeAirway1 (p=0.005)
FOXRED1 | 223128_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6489)
FOXRED1 | 223128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4372)
FOXRED1 | 223128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4442)
FOXRED1 | 223128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8083)
FOXRED1 | 223128_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FOXRED1 | 223128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0067
p=0.4198
p=0.1962
p=0.7347
p=0.3634
p=0.1366
p=0.5662
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2